

Abhinav Tripathi

EDUCATION

United College of Engineering and Research

Bachelor of Technology, Computer Science and Engineering(AIML) ;

Prayagraj, India

Dec 2021 – Sept 2025

New R.S.J. Public School

Intermediate, PCM + IT; Grade 89%

Prayagraj, India

Sept 2020 – July 2021

EXPERIENCE

Software Development Engineering Intern

Jan 2026 – Feb 2026

Xartup Fellowship

Remote

- Engineered an automated content enrichment engine using TypeScript and NLP that extracts metadata and improves website searchability.
- Implemented Web Scraping and Live AI Content Enrichment.

ACADEMIC PROJECTS

ComplexWord explorer(LexiFinder)

2023 - 2024

Python, Tkinter, NLP

- A tool that scans PDF documents for complex words, extracts them, and provides clear, concise definitions.
- Helping users to better understand and engage with the content for enhanced comprehension.

Campus Career Predictor

2024

Python, Scikit-learn, Pandas, Numpy, Matplotlib and Seaborn, Streamlit.

- Developed a job placement prediction model utilizing machine learning algorithms to forecast a student's likelihood of securing employment.

Space Debris Detection and Tracking

2024

Python, GDBT, Pandas, Numpy, Matplotlib and Seaborn, Streamlit

- Developed a web application that detects and classifies orbital debris using Two-Line Element (TLE) data. Implemented a TLE parser and engineered orbital features (e.g., apogee, perigee, velocity) using physics-based formulas.
- Integrated a trained LightGBM model for real-time debris classification.
- Designed an interactive interface with Streamlit supporting TLE/CSV uploads, visualizations, predictions, and feature importance analysis.

ACHIEVEMENTS

Programming & Problem Solving

- Solved over 150 programming challenges across various coding platforms.

5-Star Rating in C and C++(HackerRank)

- Rated 5-star on HackerRank in C and C++.

Academic Excellence

- Achieved highest marks in Physics ,Chemistry and Mathematics in HSC at school level.

CLASSES

Courses: Data Structures, Analysis of Algorithms, Database Management Systems, Computer Networks, Operating Systems, Theory of Computations

SKILLS & INTERESTS

Skills: C++, Python, JavaScript, HTML, CSS, Artificial Intelligence, SQL, GitHub, Critical thinking, Data-driven decision making

Interests: Gaming, Yoga, Cricket

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